Problem 2 | Problems T4_PhaseDiagrams I.pdf #2 | Pb-Sn · hypereutectic

Primary and Total β in a Pb-Sn Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf

2. From 2.8 kg of a Sn-Pb alloy, is it possible to have 2.21 kg of primary β and 2.53 kg of β total at (183 – ΔT) °C? Justify your answer.

Official Solution: Yes, for an alloy with composition $C_0 = 90\%$ B

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium cooling** with the Pb-Sn diagram of the problem sheet (Problems T4_PhaseDiagrams I.pdf, p. 1): eutectic at $C_e = 61.9$ wt% Sn and 183 °C; at 183 °C the α limit is $C_\alpha = 19.2$ wt% Sn and the β limit $C_\beta = 97.5$ wt% Sn.
2. **Primary β** exists only in a **hypereutectic** alloy ($61.9 < C_0 < 97.5$ wt% Sn): it is the β that forms in the β + L region before the eutectic reaction [Session 7 Slide 31].
3. Just below the eutectic temperature the primary β keeps the amount it had just above it, and the liquid present just above becomes the eutectic [Session 7 Slide 32].
4. "% B" in the key means wt% Sn (B is the Sn-rich side of the diagram).

Data

Quantity	Value
Mass of alloy m	2.8 kg
Mass of primary β $m_{\beta'}$	2.21 kg
Mass of total β m_β (at 183 – ΔT)	2.53 kg
C_α, C_e, C_β at 183 °C	19.2, 61.9, 97.5 wt% Sn

Unknown: the composition C_0. There is one unknown and two conditions, so the question is whether **both** masses can be met by the same C_0.

PHASE 2 · FRAMES & STRATEGY

The frame is the Pb-Sn diagram, composition in wt% Sn and temperature in °C; the work is on the horizontal tie lines just above and just below 183 °C, so it is one-dimensional and no change of basis is needed.

Strategy. The primary β fixes C_0 through the lever rule on the short tie line between the eutectic liquid and β (microconstituents). The total β is then **predicted** with the lever rule on the long tie line between α and β (phases), and compared with 2.53 kg. If they agree, the answer is yes.

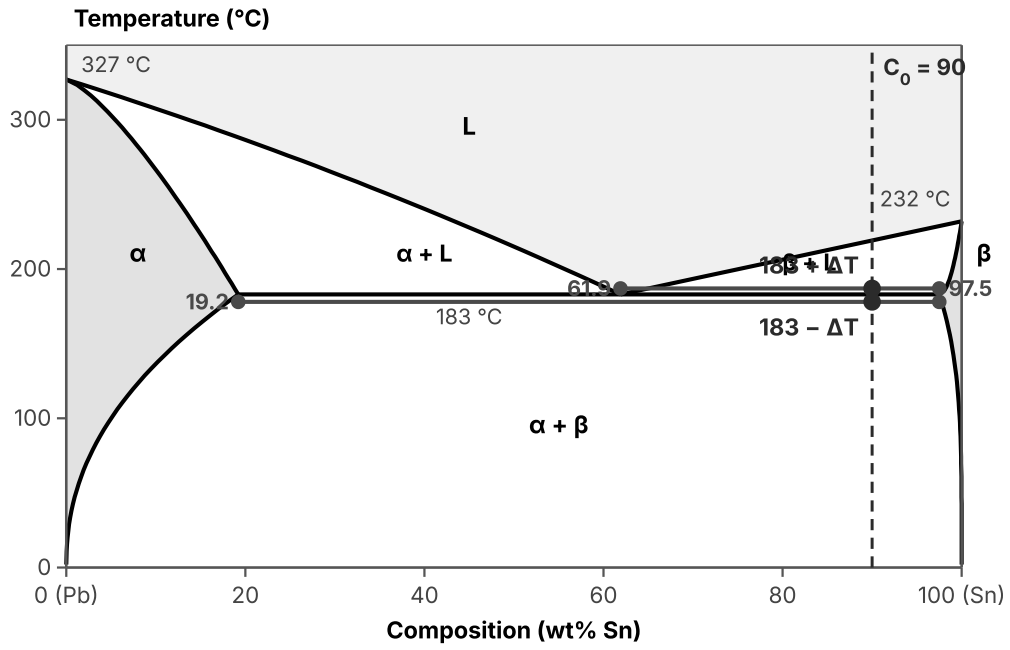


Figure 2. Pb-Sn diagram of the problem sheet (eutectic 61.9 wt% Sn at 183 °C, α 19.2 and β 97.5 wt% Sn). The alloy with 79 % primary β is the hypereutectic Pb-90 wt% Sn; the tie lines just above and below 183 °C give the primary β and the total β.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Mass fraction of primary β

$$W_{\beta'} = \frac{m_{\beta'}}{m} = \frac{2.21\ \text{kg}}{2.8\ \text{kg}} = 0.7893$$

2. Composition of the alloy Just above 183 °C the alloy is β (97.5 wt% Sn) plus liquid of eutectic composition (61.9 wt% Sn). The lever rule gives the β fraction, which is the primary β [Session 7 Slide 32]:

$$W_{\beta'} = \frac{C_0 - C_e}{C_\beta - C_e} = \frac{C_0 - 61.9}{97.5 - 61.9} = \frac{C_0 - 61.9}{35.6}$$

Solving for C_0:

$$C_0 = C_e + W_{\beta'}\,(C_\beta - C_e) = 61.9 + 0.7893 \times 35.6 = 61.9 + 28.10 = 90.0\ \text{wt\% Sn}$$

$61.9 < 90.0 < 97.5$: the alloy is hypereutectic, as primary β requires.

3. Total β predicted for this alloy Just below 183 °C the phases are α (19.2 wt% Sn) and β (97.5 wt% Sn):

$$W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{90.0 - 19.2}{97.5 - 19.2} = \frac{70.8}{78.3} = 0.9042$$

$$m_\beta = W_\beta\, m = 0.9042 \times 2.8\ \text{kg} = 2.532\ \text{kg} \approx 2.53\ \text{kg}$$

4. Answer Both conditions are met by the same alloy: **yes**, it is possible, with a Pb-90 wt% Sn alloy.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Eutectic part:** mass of eutectic $= 2.8 - 2.21 = 0.59$ kg; β inside it $= 2.53 - 2.21 = 0.32$ kg, a fraction $0.32/0.59 = 0.54$. The eutectic of the sheet diagram has $W_{\beta,e} = (61.9 - 19.2)/(97.5 - 19.2) = 42.7/78.3 = 0.545$ of β, and $0.545 \times 0.59 = 0.32$ kg: consistent.
- **Units:** masses in kg, fractions dimensionless, compositions in wt% Sn.
- **Limiting cases** (fractions dimensionless, compositions in wt% Sn): as $C_0 \to 61.9$ the primary β tends to 0 (all eutectic), and as $C_0 \to 97.5$ to 1 (all primary β); 90.0 wt% Sn lies close to the β end, consistent with 0.79 of primary β.

Interpretation The two masses are not independent: once the primary β fixes the composition, the total β is fixed too. A different total β (say 2.40 kg) would have been impossible for any alloy.

Check against the official solution

The official key ("Yes, for an alloy with composition $C_0 = 90\%$") agrees. The worked page (Solution problems Phase diagrams I.pdf, p. 6) finds $C_0 = 90\%$ from the primary β only; the check of the total β (2.53 kg) above completes the justification. The page writes the fraction as "079"; it is 0.79.